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FIRE & LOSS PREVENTION AND SECURITY REQUIREMENTS

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TRANSMISSION ENGINEERING STANDARD

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1.0 PURPOSE AND SCOPE

1.1 This Standard provides the guidelines for fire & loss prevention and security requirements in the design of Transmission Substation facilities.

1.2 The objectives of this Standard are:

1.2.1. To specify the fire and loss prevention requirements of Transmission Substations and incorporate the fire protection systems, fire alarm and detection systems applicable to Transmission Substations.

1.2.2. To incorporate safety and security requirements in the overall building design of Transmission Substations and facilities.

1.2.3. To specify the security requirement of the Transmission Substations with the minimum requirements for the provision of integrated security system applicable to Transmission Substations per SEC-05.

2.0 DESIGN CONSIDERATIONS

2.1 General

The design of a substation shall incorporate and ensure the safety, operation and protection features of the substation, such as substation shielding, insulation coordination, surge protection, relaying, grounding, fire protection system, fire detection and alarm system, security systems, lightning protection system and other protective systems and devices to ensure the safe, secure and reliable operation of the facility.

All items referred and including items not covered in this standard, shall be in accordance with the latest revision/ammendments of applicable National Grid-SA Standards and Specifications, HCIS (SEC & SAF) regulations and the latest edition of applicable International Industry codes and standards.

2.2 Security and Safety Practices

2.2.1 General Requirement

a. The classification of Security Directives (SEC) facility shall be as specified in the SOW/TS of the project. Substation facility classification shall be according to classification Criteria as attached in Appendix III. The classification of substation facility and the HCIS requirements shall be according to Turn on Letter (TOL) as specified and defined by Network Planning Department (NPD). All Substations shall be considered unmanned area unless specifically specified in the scope of work.

b. The pre-requisites for security project execution and workflow shall be based on the requirements as defined in SEC-01of High Commission for Industrial Security (HCIS).

c. Consultant, designers, suppliers, contractors and systems for security related projects shall be approved by HCIS and prequalified by National Grid Saudi Arabia.

d. Detailed vulnerability assessment or risk analysis shall be developed by a Qualified Security Consultant based on the classification of facility as specified in the SOW/TS of the project. The detailed and complete security design package prepared by the Qualified Security Consultant shall be submitted to HCIS in hard copies for approval.

e. The design considerations of the Security Systems for project implementation shall be developed by a Qualified Security Consultant and shall be according to Section 5.0 of SEC-01 of HCIS as follows;

- Base Design (10%): Shows the classification of the facility, the risk analysis and proposed security design overview.

- Detailed Design (30%): Shows details of all aspects of security related elements at facility and compliance list showing compliance or non-compliance, with justification and reasons for each applicable aspect of the security directives.

- Final Design (60% / 90%): Final Design of facility after all internal and HCIS review and Final compliance list.

f. The detailed and complete security design package shall be submitted to HCIS by ISD at least three (3) months before the initiation of the security project.

g. The security system maintenance, support and training requirements shall be as defined in SEC-01 of HCIS.

h. National Grid Saudi Arabia shall ensure that all security installations are supported with complete maintenance and support infrastructure and that the system contractor provide all details and tools required for system maintenance such as maintenance manuals, software diagnostics, special tools, calibration equipment and procedures.

2.2.2 Physical Layout

a. Pipes, pull boxes, stands and operating platforms shall be located in areas not normally used as paths by operators.

b. Provision for a secured clear zone shall comply with SEC-02 of HCIS, in addition to approved area/space allotted to substation main and support services structures.

2.2.3 Safety and Maintenance Clearances

Safety and maintenance clearances for electrical equipment shall be per TES-P-119.08.

2.2.4 Lighting and Receptacles

Exit and emergency lighting, security fence and security lightings, gatehouse, check points, yard lighting in the substation and lighting accessories, power receptacles shall be designed and installed to comply with TES-P-119.25 & SEC-01, SEC-04, SEC-07 of HCIS.

2.2.5 Fencing and Boundary Wall

a. Substation perimeter fence and boundary walls shall comply with the classification of facility as specified in the SOW/TS of the project and conform to the requirement of SEC-01 of HCIS.

Security perimeter fence shall be constructed, fabricated and installed to comply with SEC-02 Fence of HCIS.

b. Warning signs "Danger High Voltage with Bones and Skull" per standard drawing TD-80008 and "Danger High Voltage Keep Away" per standard drawing TD-80007 shall be installed in each periphery of substation fence/boundary walls and gates. The warning sign shall be fixed at an interval of 36 meters and whenever the length is more than 36 meters in any side, the warning signs shall be fixed more than one at equidistant.

2.2.6 Shielding

To minimize the exposure of direct lightning strokes on substations and substation equipment, shielding shall be provided per TES-P-119.07.

2.2.7 Grounding

A permanent protective grounding system designed per TES-P-119.10 shall be installed in the substation. All equipment and non-current carrying metal parts of the fire detection and alarm system shall be grounded. Insulated grounding conductor shall be colored green with yellow stripes with size per TES-P-119-10.

2.2.8 Fault-Sensing and Interrupting Devices

The proper relaying or fault-sensing devices in combination with interrupting devices shall protect all circuits and equipment. The combination of devices used shall isolate the fault before any further and more serious problems occur.

2.2.9 Marking and Identification of Equipment

All fire fighting equipment and facilities, hydrants and associated piping, substation electrical and mechanical equipment including auxiliary devices shall be properly marked, labeled and/or color coded for correct identification. Color coding where applicable shall be in accordance with TES-H-107.02. The space behind the wall mounted portable fire extinguishers shall be painted with post office red color.

2.3 Security Parameters

The implementation of security, safety and fire protection plans for industrial facilities kingdom-wide are the responsibility of HCIS being part of the Ministry of Interior.

The minimum requirements, limitations and application with respect to selection of material and equipment, design and construction for protection of industrial facilities shall comply and meet the latest edition of the HCIS references listed in Appendix I for each Security Directives (SEC) and Appendix II for each Safety and Fire Protection (SAF) Directives.

Industrial Security Performance Department shall be responsible to obtain all necessary approval of requirements from HCIS prior to implementation of security, safety and fire protection related projects.

3.0 SUBSTATION BUILDING

3.1 Location

All the substation building shall be located such that sufficient clearances as recommended in NFPA 70, SAF-03 & SAF-06 of HCIS are maintained between the potential fire risks (oil-filled transformers, oil-filled circuit breakers, neighboring plants, etc.) and substation building. Any credible fire from any potential fire risks shall not involve or affect the operation of equipment within the substation building.

3.2 Construction

3.2.1 General

The substation buildings (control buildings, switchgear buildings) shall be of fire resistant construction equivalent to Type I-(443) as defined by ANSI/NFPA 220 and taking into consideration SAF of HCIS regulations.

Each room inside the substation such as control room/void space, GIS room, Switchgear room/cellar, Battery room, Telecom room/void space, SCADA room, etc. shall be considered as fire compartments.

3.2.2 Substation Design Codes

The substation shall be designed and constructed according to classification of facility as specified in the SOW/TS of the project and shall conform the HCIS references listed in Appendix I for each Security Directives (SEC) and Appendix II for each Safety and Fire Protection (SAF) Directives.

3.3 Fire Resistance Ratings

3.3.1 Fire Rated Walls or Partitions

Fire rated walls or partitions (walls designed to divide an area or building into fire zones separations/fire-compartments for the purpose of containing the spread of smoke or fire within the building) shall have a minimum fire resistance rating of 3 hours as established in accordance with the test procedure of ANSI/NFPA 251 and taking into considerations the requirement of local regulations SAF-01 & SEC-01 of HCIS.

3.3.2 Fire Rated Room Doors

Fire door assemblies shall comply with the requirements of ANSI/NFPA 80 / UL 63 and SAF-01 of HCIS. The fire resistance rating of fire doors shall be certified by fire tests conducted by an accredited test agency acceptable to National Grid Saudi Arabia such as the Underwriters Laboratories Inc. and Factory Mutual Research Inc. The fire resistance of fire doors shall be classified in accordance with ANSI/NFPA 80 and shall take into consideration the requirement of the local regulations SAF-01 of HCIS. The fire door assembly shall have the same fire resistance rating (3 hours) to that of the building walls. Fire doors shall be provided with heavy duty door closers of approved type.

3.3.3 Security and Emergency Exit Doors

Security doors and emergency exit doors in substation buildings shall comply with the requirements of SEC-01, SEC-09 , SEC-10 & SAF-01 of HCIS regulation and shall have the same fire resistance rating (3 hours) to that of the building walls. Emergency Exit Doors shall be provided with Heavy Duty Door Closers of Approved Type.

3.3.4 Fire Stops and Seals

a. All penetrations through fire rated walls or partitions, floors, ceilings, HVAC ducts and exterior walls by piping services, electrical cables, cable ducts, conduits and similar installations shall be sealed by approved materials or fire stops with a fire resistance rating (3 hours) equal to or greater than that of the floor, wall or ceiling.

b. All penetrations and conduits shall be sealed by approved material to prevent the transfer of smoke, flammable gases or vapors from one area to another.

c. Fire stops used for cable penetrations through fire rated floors, walls or partitions shall meet the requirement of ASTM 3814 or UL 1479 and taking into consideration on the requirements of NFPA 221 and the local regulations SAF-01 of HCIS.

3.4 Equipment Layout

The physical arrangement of equipment housed within a substation building shall be in accordance with the guidelines of TES-P-119.19 and taking into consideration the requirement of local regulations SAF-01 of HCIS, with emphasis on equipment clearances, working space, maintenance space and ventilation. The guidelines on the installation of substation equipment established in Part 1 of the National Electrical Safety Code, ANSI C2, and in Part III of the Uniform Mechanical Code (UMC) shall be observed where applicable.

3.5 Heating, Ventilating and Air Conditioning (HVAC) System

3.5.1 Duct System

The HVAC duct system shall comply with the applicable requirements of NFPA 90A**,** SMACNA (Sheet Metal and Air Conditioning Contractor's National Association), related specifications and drawings. The HVAC ducts, air terminal devices, accessories and supports shall be constructed of non-combustible or fire resistant materials.

Duct insulation and insulation adhesives shall have a flame spread rating of not greater than 25 and a smoke developed rating of not greater than 50 when tested in accordance with NFPA 255.

3.5.2 HVAC Equipment Shutdown

a. The HVAC control system shall be designed and operated to receive an input signal from Fire Alarm Control Panel to shutdown the HVAC system & to close the motorized fire /smoke dampers associated with the operation of the Extinguishing system as per TES-B-106.01, TES-K-100.01 and SAF-01 of HCIS local regulations.

b. Approved multi-sensors (combined optical and heat detectors) / photoelectric types duct smoke detectors with built in sampling tubes shall be installed in both the main supply and return ducts of the air handling unit to initiate signals for the automatic shutdown of the air conditioning unit and its smoke/fire dampers.

c. The operation of the duct smoke detectors shall be supervised by the fire protective signaling panel. For non-ducted return air system (plenum type) the return air duct detector will be omitted provided that sufficient adequate number of ceiling-mounted smoke / heat detectors are installed in the protected room and /or compartment.

3.5.3 Fire Dampers

a. All HVAC ducts penetrating fire rated walls, floors or partitions shall be provided with UL listed/labeled fire dampers in accordance with ANSI/NFPA 90A.

b. For protection of openings in walls, partitions or floors with fire resistance ratings of less than 3 hours, fire dampers shall have 1-1/2 hours fire protection rating.

c. For protection of openings in walls, partitions or floors with fire resistance ratings of 3 hours or more, fire dampers shall have a minimum 3 hours fire protection rating.

d. In substations, motorized combination fire and smoke dampers shall be installed in the main supply and return duct penetrations through building wall, branch supply and return duct passing through partition walls, transfer air grilles and pressure relief dampers on the room partition wall(s) protected with Fire Suppression System. Motorized combination fire and smoke dampers shall comply with UL555 and UL555S.

e. All other duct penetrations and openings for louvers and grilles in fire rated walls and partitions shall be provided with fusible link operated fire dampers with fusible link rating of 71˚C. Fire damper shall comply with UL 555.

f. The duct penetrations shall be sealed on both sides of the wall with 10 gauge galvanized steel sheet metal enclosure.

3.5.4 Removal of Smoke After Fire

a. Natural ventilation, Exhaust fan, Smoke Purging fan for the removal of products of combustion shall be provided in control rooms, telecom room, clean agent room, MV switchgear rooms, GIS room, Cable Basement or Cable Entry Room and Battery room.

b. Panels on sky window opening or any other approved method of vent opening under the protected room/facility shall be arranged to be opened remotely from safe and pre-approved location. Such windows and panel and their control devices shall be clearly identified with approved type label that are fabricated and written in Arabic and English text.

c. Smoke purging fan shall be provided to forcibly vent the smoke, extinguishing discharged gas from the substation enclosures, with a selector switch. The selector switch shall be installed at a readily accessible location outside the protected area/facility. Selector switch labels shall be provided that read as follows and shall be written in Arabic and English text.

(Arabic Text) OFF Position - Normal Setting at Normal operating status of the system.

(Arabic Text) ON Position - To be set only by concerned personnel after a fire incident or when extinguishing period is reached.

d. The capacity of smoke purging fan shall be calculated based on latest ASHRAE air dilution (air changes per hour) as mimimum requirements using the empirical formula: Capacity of fan required = Volume of room(s) (LxWxH) x number of air changes per hour (m^3/hour).

One unit of purging fan shall be provided as per HVAC portions specified in SOW/TS or as per air conditioning system approved design or groups such as below:

- Group I (GIS room and its basement)
- Group II (MV Switchgear rooms and its basement)
- Group III (Control room, telecom room and clean agent room)

3.6 Emergency Exits

The substation building emergency exits shall be designed in accordance with NFPA 101. All hazardous areas in the substation such as cable basement / entry room & cable tunnels, at least two (2) exits (exit stairs) shall be provided but in no case the travel distance to the nearest exit shall not exceed 23 meters.

Doors for emergency exits shall comply with the requirements of clause 3.3.2 or clause 3.3.3 as applicable.

Exit staircase shall be provided if HVAC units are installed at top of roof without any provision of vertical ladder.

Vertical ladder with encasement shall be provided if no HVAC units are installed at roof top.

3.7 Fire Exit Hardware

Any door as a means of egress including door in the battery room shall be fitted with UL listed/labeled fire exit hardware as per the requirements of NFPA 80 and NFPA 101.

3.8 Fire & Safety Requirements

3.8.1 General Requirements for Executing Fire Protection Projects

The pre-requisites for fire protection projects execution shall be based on the requirements as defined in SAF-01of HCIS.

Designers, suppliers, contractors and systems for related projects shall be approved by HCIS.

National Grid Saudi Arabia main contractor, in consultation with qualified security consultant, shall be responsible for preparation of preliminary design package that details the scope of work, main and auxiliary elements of safety and fire protection related work and site specific requirements to be submitted to HCIS for review and approval prior to commencing detail design, bidding and contract award for design and installation.

The preliminary design package shall include the following minimum information:

a) Site plan view drawings showing the layout of the safety and fire protection related devices and systems relative to the protected facilities.

b) Areas covered by the fire protection/detection system.

c) Specifications for fire protection/detection equipment at each site.

d) Location and installation details of end devices

After the approval of the preliminary design package by HCIS, detail design package with complete detailed technical specifications, engineering design and list of bidders for all elements shall be prepared and the complete hard copies of this package shall be submitted to HCIS by ISD for approval at least three (3) months prior to project execution.

The detail design package to be submitted to HCIS shall include the following:

- List of contractors on the bidders list
- Certification of the contractors from Ministry of Interior (Civil Defense) and Ministry of Commerce and Industry.
- List showing non-compliance, with justification and reasons, for each applicable aspect of the HCIS safety and fire protection diretives and project specifications, if any.
- List of major equipment, systems, their suppliers and their local agent
- Proposed work execution schedule
- Copies of original manufacturer catalogs and device specifications.
- Accurrate disclosure of full identities of all sources and manufacturers technologies, devices and systems in the documentation

The progress report shall be submitted to HCIS every six (6) months on the progress of major safety and fire protection related projects.

3.8.2 Substation Safety Requirements

All substations buildings having a capacity of 110kV, 115kV, 132kV, 230kV and 380kV are considered unmanned. These are the building facilities that are designated to house the sensitive equipment that shall be provided with the following fire, safety & security structure to comply with governing regulations for the provision of the following:

- Fire Detection and Alarm Systems (FDAS)
- Fire Alarm/Fault Signaling Panels
- Fire Alarm / Fault Data Printer
- Fire Suppression System & Equipment
- Automatic Water Fire Extinguishing Systems
- Portable & Fire Extinguishers
- Security Systems
- Safety and First Aid Equipment
- Civil Structural / Passive Safety Parameters

3.8.3 Fire Detection and Alarm Systems (FDAS)

a. Substations are normally compartmented to a pre-designed individual service functions rooms. These rooms / areas that are in the facility shall be monitored with Automatic fire detection and alarm systems. The fire alarm systems shall provide for an extensive detection of fire, smoke, flame and heat in the substation building to cover the following areas:

i. Office
ii. Kitchen or Tea Room
iii. Workshop/Record Room
iv. Store room
v. Hallway, Corridor, Lobby & Stair cases
vi. Lobby
vii. Oil Filled Transformer unit
viii. GIS Room (69kV to 380kV voltage ratings)
ix. Switchgear Rooms (13.8kV to 34.5kV)
x. Communication Room
xi. Cable Entries, Basement and cable tunnels
xii. Battery room
xiii. HVAC Equipment (Mechanical) Room
xiv. HVAC Main Supply and Return Ducts
xv. AC/DC Distribution Room
xvi. False Ceilings
xvii. Floor Voids
xviii. Cable Trenches
xix. Fire Pump Room
xx. Auxiliary Transformer / RMU Area
xxi. Clean Agent Cylinder Room for Gas Extinguishing System
xxii. Any exceptions to the above requirements shall qualify for exception in accordance with NFPA 72

b. The types of detecting and actuating means for all areas including the support services rooms shall be according to requirement specified in Table 1.

c. The fire Detection and Alarm System shall be served with a dedicated control panel that is physically and functionally separated from extinguishing system provided. The status condition of the system shall be arranged to be annunciated remotely through the local SCADA Marshalling Panel.

d. The Fire Detection and Alarm System shall be designed and installed in accordance with TES-B-106.01, NFPA 72 and taking into consideration the requirements of local regulation SAF of HCIS.

e. The system shall be complete with fire alarm control panel, automatic fire detectors, manual fire alarm stations, visual and audible alarm indicating devices, associated circuits, data printer, warning signs and ancillary equipment. A mimic panel shall be provided beside the Main Fire Alarm Control Panel located in the control room. Mimic panel shall show the actual site layout with each zone represented by a red LED.

f. All materials and equipment of the fire detection and alarm system shall be listed (UL/FM approved).These materials shall be selected from National Grid Saudi Arabia approved manufacturer's list.

g. The Fire Detection and Extinguishing System shall operate from a mains power supply unit with required DC voltage & current output with adequate back-up battery system, which shall be capable of emergency operation for at least 48 hours and 30 minutes of alarm condition.

h. The automatic detectors shall be of plug-in type with separate base and sensing unit. LED type lamp shall be provided with separate base where and when required for which shall operate when the detector has been activated. The lamp shall remain "ON" until the system has been completely reset.

i. LED provided for ceiling void shall be installed in the suspended ceiling and for floor void a mimic panel shall be provided on the wall beside the main door inside the room with each detector’s location clearly identified as per TES-B-106.01.

j. Care should be taken to protect the detectors from close proximity of air flow from ducts. Detector shall be installed not less than one meter (1m) from air diffuser as per NFPA72.

k. The type of detectors used for the fire protection system shall be as per TES-B-106.01, adequate number of detectors shall be provided in each room, floor and ceiling voids.

l. Weatherproof Rate Anticipation Heat Detectors for transformer protection shall be fixed temperature type. Detector Rating shall be as per NFPA 72 latest edition requirement.

m. For other areas, fixed temperature heat detectors shall be entirely electronic in operation and have a thermistor as ambient temperature sensor to be fully resettable. Detector temperature rating shall be approximately 57 ºC.

n. When Combination of Fixed-Temperature and Rate of Rise Heat Detectors are used the fixed temperature element shall be independent of the rate of rise element. The rating for the fixed temperature element shall be 57 ºC.

o. All elements of detectors shall be resettable and entirely solid state. Adequate special detectors as required for air-conditioning system shall be provided accordingly. Duct detectors shall be assigned /wired in a separate zone.

p. The fire and smoke in the areas described in clause 3.8.2a shall be detected by a combination of detectors as explained in Clause 6, Table 1. Fire alarm system equipment located in the battery room shall be explosion proof and suitable for classified (hazardous) locations (Class 1, Division 2, Group B "Hydrogen" atmosphere).

q. Smoke detectors installed at the switchgear room (69kV to 380kV) shall be located such as to allow safe maintenance while the switchgear is energized.

r. Automatic fire detectors shall be addressable type and suitable for wiring in class A circuit. It shall be intelligent/ diagnostic type and compatible type with the microprocessor based control and indicating equipment and shall comply with the requirements of TES-B-106.01. Automatic fire detectors shall be self-protected against electro-magnetic interference for protection against false alarm. Each detector can be isolated from the system for maintenance purposes without affecting the entire zone.

s. Manual alarm call points shall be provided at several convenient points located in the natural path of exit, so that alarm can be given by operating personnel independently of the automatic system. The travel distance for manual call point shall not be more than 30m. These shall be Push & Pull Station type and the word ‘FIRE ALARM’ in Arabic & English engraved. The color of these alarm points shall be Red. Manual call point for a particular zone shall be connected on the same

circuit as the detector being provided for the zone. Mounting height shall be 1.4m from finish floor level.

t. Manual fire alarm stations shall also be installed at all basement stairs, exit doors. The manual stations shall be addressable/diagnostic type. The same shall be weatherproof, when installed outdoors, manual fire alarm devices shall be installed in all areas, whether or not the area is provided with automatic extinguishing system.

u. Indoor audible alarm devices shall be dome type alarm bells, minimum 150mm diameter gong, with strobe light finished in red, installed at a height of 2500mm from the finished floor level or 300mm below false ceiling and shall be capable of producing continuous ringing as required. At least one alarm bell shall be provided per fire compartment. Bells shall be distributed and connected in at least 2 alarm circuits. The sounders at any point in the building should produce a minimum sound level of of 85 dB measured at 3000mm distance.

v. One weatherproof flasher having diameter not less than 100 mm shall be installed on a 1.5 meter pole above the highest portion of the building of the substation and facing in front of main road. It shall have red flash light visible enough to be noticed even during day time for identification by the Civil Defence Force. However, the height of the flasher shall be less than that of lightning arrester. The flasher shall get actuated from fire protective signaling panel on the actuation of fire detection.

w. All wiring for fire protective signaling systems shall comply with Article 760 of the National Electrical Code, NFPA 70 and ANSI/IEEE C2. Acceptable wire systems are:

i. Listed /approved Fire Alarm rated 105 ºC shall be provided for addressable system.

ii. Heat resistant cables, preferably, THW/THNN stranded shall be used for the extinguishing system wiring utilizing conventional scheme.

iii. The wire size shall be as follows:

- 1.5 mm² for signal initiating devices.
- 2 mm² for alarm devices.
- 2.5 mm² for power supply.

Refer also to the specification 11-TMSS -10 for conductor requirements.

iv. The wiring system shall be in galvanized Rigid Steel Conduit (RSC) when service mounted or in Polyvinyl Chloride (PVC)

conduit when embedded in wall. Electrical Metallic Tubing (EMT) shall not be used.

x. All wiring for fire protective signaling systems running outside the building shall be installed in National Grid Saudi Arabia approved direct-buried conduit.

y. Fire protection miscellaneous details shall be according to TSD TA-800106.

3.8.4 Fire Alarm/Fault Signaling Panel

Refer to 90-TMSS-01 for full specification.

a. Remote Signaling Panel

The fire protective signaling panel shall have the provision for remote signaling of fire alarm/fault signal through an interface connection with SCADA RTU / IFC provided in the substation.

The fire alarm system shall also be provided with a tele-alarm device (automatic telephone call transmitter) that will facilitate the automatic transmission of fire alarm signals through the installed substation telephone line. The tele-alarm device (automatic telephone call transmitter) shall be designed and manufactured for heavy industrial applications, provided with a digital voice and data messaging system, and will provide for automatic repeating of alarm up to five locations through an internally incorporated automatic telephone number dialer. Additional details shall be coordinated with National Grid Saudi Arabia.

Any fault in the system shall be shown on the panel by means of an amber light.

3.8.5 Fire Alarm/Fault Data Printer

400/230Vac data printer shall automatically print all incoming alarm and trouble status and faults received by the fire protective signaling panel with date and time of receipt. The printer shall be capable to print 264 characters per second (cps), 80 columns. A steel enclosure with lockable, clear glass-front per Figure 21-1 shall house the printer and paper. The printer and paper shall be placed on a withdrawable shelf inside the enclosure for easy access.

3.8.6 Fire Suppression System & Equipment

Substation has an inherent electrical fire-loads that could be quantified and classified as deep seated fire. The cost of material and the service losses to the end-users could be huge in case of fire accident broke out.

The impact of material and service losses could be considerably be reduced with the provision of approved type fire fighting systems. Appraisal on insurance value of the facility will be of higher return and consequently of less insurance premium paid.
The type of fire suppression system and the extent of the protection to be provided shall depend on the following considerations:

- Value of the protected equipment
- Safety to operating personnel
- Nature of fire hazard
- Compliance with SAF-04 of HCIS, local and international safety codes and regulations
- Criticality of one station for system operation

The type of clean agent or chemical agent to be used shall be as specified in the SOW/TS of the project.

The recommended fire fighting systems for the protection of substation facilities are as follows:

3.8.6.1 Clean Agent Extinguishing System

The total Flooding of Clean Agent Extinguishing System shall be UL Listed , designed and installed to comply with TES-B-106.05, TES-B106.06, NFPA 2001 and the requirements of the local regulation SAF of HCIS.

The areas listed below that shall be provided with clean agent extinguishing system:

a) MV Switchgear room
b) Control room
c) Communication room
d) SCADA room
e) GIS LCC panels
f) AC/DC room
g) Battery room
h) Indoor type transformer (if any)

The minimim clean agent design concentration to be used shall be based on the latest edition of NFPA 2001.

The Clean Agent shall be achieved in the vicinity of all equipment or area to be protected within one minute after the end of the initial discharge and the concentration shall be maintained with no appreciable dilution for at least 10 minutes. In addition to the concentration requirement, additional quantity of agent shall be provided to compensate for openings that could not be closed, forced ventilation or other specified conditions, which would affect the extinguishing efficiency (All design factors shall be as per the

requirement of NFPA 2001). The total quantity of Clean Agent shall be calculated from the risk dimension including floor voids and ceiling voids.

Cylinders, manifolds, pipes & all other system components shall be installed by an approved Contractor. Complete system shall be installed and fixed securely as per manufacturer's instructions & NFPA 2001 requirements.

Clean Agent cylinder valves shall be of differential pressure design and not require an explosive / consumable device to operate them. Solenoid valve shall be of re-settable type. Disposable type is not acceptable.

Manual operation of the system shall be possible through the extinguishing control panel. All clean agent cylinders shall be located in the cylinder room. For multiple systems designed in large areas, manual mechanical/pneumatic actuator shall be provided at the main entrance of the protected area for simultaneous discharge of all systems' cylinders.

Manual Actuator with handle shall be provided on the agent cylinder valve in case non actuation/failure of Electric Actuation.

Manual Release Unit shall be provided on the wall beside the main entrance to each protected area.

Motorized fire dampers operating on the first alarm shall be provided to seal off room openings, i.e. air return ducts or exhaust, if any in conjunction with the operation of the detection system.

Clean Agent system shall be provided with 100% reserve cylinder (connected with the system piping) and accessories.

System Functional operational test shall be performed as per NFPA 2001 requirement.

Room integrity test shall be carried out for all areas protected with Clean Agent.

Clean agent cylinders shall be connected to the actuators only after the successful enclosure integrity test is achieved to prevent accidental discharge while working.

Each cylinder shall be provided with low pressure switch to monitor agent pressure drop in the cylinder. The respective cylinder shall send a signal to corresponding Fire suppression releasing panel as "Low Cylinder Pressure" and signal shall be indicated in the main panel.

Maximum capacity of clean agent cylinder shall not exceed 750 pounds. A hydraulic trolley with minimum of 750kg capacity shall be provided for each project.

Cylinders & pipes shall be fully earthed & cylinders shall be standing on rubber mat for Safety purposes.

All drawings showing the layout, riser diagram of the system and operating instructions are to be fixed beside the extinguishing system control panel in the room under protection inside glassed frame. Cross zoning method of installing detectors shall be applied. Floor void detectors shall be provided with sign on the floor tiles.

Clean Agent protected areas shall be provided with the following:

- Lock-off panel (auto-manual switch, 3-way status indicators, and manually operated dual action type release station) shall be provided on main entrance to protected area.
- Clearly visible LED type status indicator (auto-manual discharge) located above the main door in the protected area.

Each area protected with clean agent system shall have independent control panel.

A separate Clean Agent cylinder room shall be constructed, with safe access to facilitate for a possible manual and mechanical actuation of the system. The room shall be air-conditioned supplied from substation central air conditioning system. & protected with CO_2 portable fire extinguishers.

Clean Agent protected areas shall be fully compartmentalized. Motorized duct dampers working simultaneously with the first alarm of fire detection system shall be installed. Automatic door closer shall be provided.

Schedule 40 Electric Resistance Welded (ERW) to ASTM A53/GR-A Galvanized Pipes and Galvanized threaded fittings shall be provided. Grooved coupling for pipes 65mm diameter and above shall be utilized.

For floor void mounted detectors, a mimic panel shall be provided on the wall near the main door with each detector's location clearly identified.

All equipment shall be UL listed/FM approved except the following:

- Extinguishant and lock-off panel – equivalent listing required.
- Pipes and fittings – conformity with NFPA standards.

Equipment summary showing make/brand, listing/approval, quantity, made, number of equipment, etc., shall be provided. Basic diagram/layout of proposed system shall also be provided.
After installation, all system pipe work shall be blown free from swarf and debris using compressed air, nitrogen or carbon-dioxide before the system is commissioned, to verify that flow is continuous, and the piping and nozzles are unobstructed. All the tests shall be carried out in the presence ISD Reprentatives including the commissioning tests. All Clean Agent piping shall be tested pneumatically as per NFPA.

3.8.6.2 Lock off panel (Gas System Status Unit)

For rooms provided with automatically operated gas extinguishing system, Lock off panel shall be provided on the wall beside the main entrance to each protected area and shall include the following:

i. Auto-manual selector switch
Provide auto/manual selector switch to select the gas extinguishing system in auto or manual mode of operation.

Operation of this unit in manual mode shall activate false buzzer send signal to local annunciator panel as Clean Agent/gas in manual mode and to Power Control Center / SCADA as ‘Fire system Faulty’.

Instruction label shall be provided beside the unit as given below:

Prior to operation of this unit please inform Power Control Centre. Operation of this unit will send a signal of “Clean Agent/gas system in manual mode/Fire system faulty” to substation annunciator panel. Before leaving substation, ensure that this unit is in “Auto” mode and reset annunciator panel in Control/Relay room.

ii. System Status Indicator

Provide LED type system status indicator. It shall indicate the system as:

- Auto Mode
- Manual Mode
- Gas released

iii. Gas Release Unit (Electrical)

Provide electrically operated dual action pull station type, to be located outside the protected area, release unit. Operation of this unit shall bypass the system's auto mode and initiate the release of gas. Prior to gas release, all Air handling Units **(**AHU**)** / Packaged Type Air Conditioning Units (PACU) shall trip and motorized dampers closed . Moreover, all AHU and PACU shall be provided with maintenance switches for isolation when required for maintenance and/or repair.

3.8.6.3 Gas System 3- way Status Indicator

This unit shall include system status indicators with same features as mentioned in item 3.8.6.2.ii above. This unit shall be installed in a suitable location, preferably above door, inside the protected area.

3.8.6.4 Automatic Water Fire Extinguishing Systems

Fire Water System for the protection of substation facilities other than that of what is prescribed on preceding "clauses" call for the provision of the following:

- Water Spray Fixed System
- Wet Sprinkler System for Fire Pump room
- Yard Fire Hydrant Systems & Fire hose cabinet

3.8.6.5 Water Spray Fixed System

Only rate of anticipation thermal heat detector shall be provided for the following:

a. Outdoor Transformers, of complete Electrical System Protection, and passive fire Protection as indicated in clause 3.10 and in accordance with the NPFA requirements.
b. Spare transformers not intended to be used in place.
c. Transformer containing less than 500gallons (1890 liters) of combustible transformer oil.

Outdoor transformers which does not meet above conditions, Water Spray Fixed System shall be provided as per requirements of the latest edition of NFPA 15. Detection and actuation shall consist of cross-zoned rate of anticipation thermal heat detector as per system requirements of NFPA 72.

3.8.6.6 Wet Sprinkler System

Fire Pump room, shall be protected with Wet Sprinkler System. The system shall be designed & installed to comply with NFPA 13

and the requirement of SAF of HCIS. Fire Detection and Alarm system shall be Bulb type sprinkler rated 57 ºC.

Fire department connection shall be provided.

All equipment in the pump house, if available – pumps, drivers, controllers, etc. shall be waterproof.

3.8.6.7 Yard Fire Hydrant System & Fire Hose Cabinet

Substation Facility Fire Hydrant System requirements shall be defined as described below:

Substations Outdoor Switchyard having an operating capacity rating of 230kV and 380kV shall be furnished with hydrant system equipment & Fire Hose Cabinet equivalent to what is defined below:

a. Fire Pump Systems

Electric Motor Driven, Diesel Engine Driven of UL listed & Jockey Pump including UL listed separate controller for each pump shall be provided. Horizontal centrifugal type fire pump shall be used for above ground fire water supply source and vertical turbine type fire pump shall be used for underground fire water supply source.

Pressure tank shall not be used as among the fire water supply system equipment/devices.

Diesel fuel tank located outside the pump room with shed protection & shall be designed for adequate fuel capacity with the following:

- Fuel level indicator
- Filling connection
- Drainage connection.

b. Dry Type Fire Hydrant Unit

Adequate number of Fire Hydrants & fire hose cabinets shall be provided as per NFPA & SAF of HCIS requirements. Hydrant shall have 2 x 2.5 outlet with cap and chain. In no case, the fire hydrant shall be spaced more than 70 meters apart.

c. Hydrant Hose House (Red Finished)

The hydrant hose cabinet shall be equipped with 65mm diameter adjustable fire hose nozzle, pinch point crow bar, fire axe, adjustable hydrant wrench , reducer 2.5/1.5 inch and synthetic rubber lined (double jacket of cloth material) red hose 2x65mm diameter, 15 meter length with instantaneous coupling.

d. Fire Hydrant Piping Network

Piping network from fire-water supply source to hydrant outlet shall be of above ground installation. Pipeline traversing across roads shall be installed inside the concrete pipe trench with moveable concrete cover. The pipe network will be extended to all provided fire water system.

e. Fire Water Supply Tank

Fire pump house and fire-water tank shall be constructed from concrete structure. The water capacity of the tank shall be based on water demands of the single biggest fire hazard as determined in approved hydraulic calculations & complying with NFPA requirements.

3.8.6.8 Portable & Trolley Type Fire Extinguishers

Approved UL listed portable fire extinguishers conforming to 90-TMSS-03 /ISS specification 0611-005 specification for portable & wheel type fire extinguisher, NFPA10 and taking into consideration the requirement of SAF of HCIS shall be adequately provided in the substation building, but not be limited to the following areas:

a. Battery Room (CO_2 20 Lbs inside, Dry chemical 17 Lbs outside near entrance door)
b. HVAC / PACU Room and Mechanical Room (Dry chemical 17 Lbs)
c. Pump room (CO_2 20 Lbs)
d. Workshops and Offices (CO_2 20 Lbs)
e. MV Switchgear Rooms (CO_2 20 Lbs)
f. Communication Room (CO_2 20 Lbs)
g. Control and Relay Room (CO_2 20 Lbs)
h. Cable basement or Cable Entry Room and Cable Tunnels (Dry chemical 17 Lbs)
i. Security Gate House (if available) (Dry chemical 17 Lbs)
j. GIS Room (CO_2 20 Lbs)
k. AC /DC Room (CO_2 20 Lbs)
l. Hallway (CO_2 20 Lbs)

m. Power Transformer Parking Area (Wheel Type 125 Lbs & Portable type 20 Lbs Dry chemical)
n. Station Transformer Parking Area (Dry chemical 17 Lbs)
o. Shunt Capacitor Parking Area (Dry chemical 17 Lbs)
p. Shunt Reactor Parking Area (Dry chemical 17 Lbs)
q. Toilet (Dry chemical 17 Lbs)
r. Clean Agent Cylinder Room (CO_2 20 Lbs)

The number of units shall be governed by individual conditions. In general, suitable type and numbers of extinguishers shall be provided at a distance as specified in NFPA 10.

All outdoor assigned or mounted fire extinguisher units shall be supplied with housing or shade to prevent from direct exposure to sunlight.

Likewise, indoor assigned fire extinguishers units shall be mounted on walls which will be located in the pre-approved location.

Portable fire extinguishers shall be provided in all areas whether or not protected with fixed fire suppression systems.

The fire extinguishers shall be located near entrance and exit doors and along the normal path of travel. They shall be visible and readily accessible. Fire extinguishers shall be mounted on the wall with its top not more than 1070 mm. above the finished floor.

At least two (2) wheeled type minimum 125 lbs **C**lass ABC fire extinguisher shall be provided in the substation yard, where oil-filled equipment are present and the extinguisher shall be protected by a shed. Flooring under the shed shall be of concrete and the three sides of the shed shall be provided with walls made of G.I. sheets.

3.8.6.9 Security Systems

The Security Systems for the substation shall be designed and constructed according to classification of facility as specified in the SOW/TS of the project and shall conform the HCIS references listed in Appendix I for each Security Directives (SEC), the design of the Security System shall be developed by a Qualified Security Consultant. The Security System shall consist of CCTV and Security Control System equipment.

3.8.6.10 Safety and First Aid Equipment

a. First Aid Equipment

All substation buildings shall be provided with first aid box located in the control room. Posters outlining emergency treatment for electrical shock, cardio pulmonary resuscitation and artificial respirations shall be provided. One folding type stretcher shall also be provided in the control room.

b. Safety Equipment

Safety equipment and devices for personnel protection, safe handling and proper maintenance of batteries shall be provided in the battery room per TES-P-103.04 and TES-P-119.19.

c. Life Support Equipment

Life support equipment in the substation buildings shall be provided to minimize hazards to personnel under abnormal operating conditions.

Self-contained breathing apparatus of 30 minutes capacity shall be provided in the control room near the entrance/exit door and/or in rooms or areas normally occupied by personnel. It is also required that self-contained breathing apparatus be made available for the operator's use in rooms or areas protected by fixed fire extinguishing system utilizing an agent that could cause a breathing hazard.

The self-contained breathing apparatus shall be stored in a box mounted on the wall. The storage box shall be marked with "SELF-CONTAINED BREATHING APPARATUS" written both in English and Arabic languages.

d. Signs

i. Adequate fire / safety warnings in the form of signs, labels, decal placards or other markings shall be installed in all locations where it requires visual alertness against degree and nature of hazard(s) which can cause injury or death. Signs shall also be provided for precautions or evasive actions to take and directions for eliminating the hazard. All fire/safety warnings shall be per **"Safety Signs and Warning Devices Special Manual # 65-110"** issued by Industrial Security Department and shall be written in both Arabic and English.

ii. Illuminated exit sign other than the main exterior exit doors that obviously and clearly are identifiable as Exits be marked by an approved sign that is readily visible from any direction of Exit access. The exit signs must be illuminated by incandescent or fluorescent bulb fed from an Emergency stand-by DC Power per NFPA 101.

iii. Signs must be located in such a location with such size, distinctive color and design to be readily visible. All signs must have contrast with decorations, interior finish, or other signs and may not be block by decorations, furnishings, or equipment that impairs visibility of such sign. Further, no brightly illuminated sign, display, or object that could detract from an Exit sign are permitted in or near the line of vision of an Exit sign.

iv. Every sign required for Exit or Exit access other than those where operations or processes require low lighting levels must suitably illuminated by a reliable light source. Externally and internally illuminated signs must be legible in both normal and emergency lighting mode.

v. All doors, passages or stairways that are neither an exit nor a way of exit access yet are likely to be mistaken for an exit be identified with a "No Exit" sign.

vi. All Exit signs with a directional indicator be placed in every location where the direction of travel to reach the nearest exit is not apparent.

vii. All Exit signs must be of a distinctive color and be illuminated to a surface value of at least five foot-candles (54 lux) by a reliable light source.

3.8.6.11 Civil Structural / Passive Safety Parameters

a. Barrier

When the design and size of the containment facilities utilized are inadequate, it may be necessary to install some form of barrier to protect other substation equipment or neighboring properties. These barriers shall be totally constructed of non-combustible materials, such as concrete blocks, bricks, steel sheets and reinforced concrete, and designed to withstand the thermal shock of the largest credible fire to which they may be subjected.

b. Doors

Doors shall be UL listed, Swing type metal fire door shall be 3 hours fire rated.

c. Structures and Supports

Structures supporting disconnect switches shall be kept as low as possible when they are placed near oil-filled equipment to avoid the extreme high temperatures caused due to fire at the upper levels. Any essential structure or support that may be subjected to an oil fire shall be constructed of steel rather than aluminum.

d. Cable Penetration Fire Stops, Fire Breaks and Coatings

i. Cable Penetration Fire Stops

Cable penetration fire stops shall be installed to prevent fire propagation along a cable system through a fire-rated wall, floor, or floor-ceiling barrier while maintaining the integrity of the fire barrier through which the cable system penetrates.

ii. Cable Fire Breaks

When the cables used do not meet the flame propagation requirements of ANSI/IEEE 383, fire hazard can be minimized by utilizing fire breaks.

Cable fire break can be achieved by coating a 3m section of the cables with approved fire retardant coatings or equivalent fireproofing tapes.

Cable fire breaks shall be installed in cable trays or open raceways at intervals not exceeding 6 meters.

iii. Cable Paint Coating

Fire retardant coating paint, white color, rated for 3 hours shall be applied to the cables.

3.8.6.12 Firefighting Equipment Physical Requirements

The firefighting equipment physical location, mounting, demarcation, paints and signs requirements shall be per Element 3.2 - Locations marked, Floor Clear of Saudi Electricity Company 5-Star Standards Safety and Health Management System Manual.

3.9 Security, Access Door Systems & Equipment

The substation facilities requirements for Security, Door systems and Equipment shall meet the minimum requirement of HCIS. Refer to Appendix l for the list of Security Directives (SEC) for industrial facilities.

3.10 Oil –filled Equipment

The presence of oil-filled electrical equipment in a substation such a transformers, reactors and switchgear equipment require special considerations for their location, spacing and fire protection.

3.10.1 Transformers

Transformers generally contain the largest quantity of combustible substance located in a substation. Therefore, special attention shall be given to their location, relaying and fire protection. Installation of oil-filled transformer inside the substation building is not permitted.

For Outdoor Transformers conditions shall be applied as per clause 3.8.6.5.

3.10.2 Outdoor Installation of Oil-Filled Transformers

The following safeguards shall be applied to the outdoor installation of oil-filled transformers. One or more of these safeguards shall be applied according to the degree of hazard involved.

a. Oil Containment

The spilled oil from a ruptured transformer tank shall be collected and confined to prevent the spread of fire to other areas or equipment.

Various types of holding pit designs are presently in use by the Utilities. The area below and surrounding the oil filled-equipment is used as a collecting pit. Pits shall be designed so that the collected oil and water will not weaken the equipment foundations. The most common designs include those listed below:

i. Oil Retention Pit or Oil Catch Basin

The foundation of the oil filled equipment is designed in the form of an open rectangular box with the oil filled equipment placed on concrete pier. The rectangular concrete box below the oil filled equipment is designed to hold 100% of the equipment oil, with the top surface of the oil at least 500mm below the top of the retention pit foundation box. The

space around the rectangular foundation box shall be used as a part of retention pit oil catch basin by providing interconnection piping to keep the depth to a minimum.

A metallic grating with wire mesh and gravel over it shall be installed at 200mm below from the top of the foundation. Alternatively, any other improved civil work shall be employed with the approval of National Grid Saudi Arabia concerned department. A sump with checkered steel cover and suction pipe is constructed within the retention pit/catch basin. Pipes from the retention pit/catch basin towards the sump shall be provided. A permanent sump pump shall be installed to remove collected rain water. The pump shall be manually operated in attended and unattended substations.

For transformer rated above 100 MVA, if the rectangular concrete box below the oil filled equipment is not designed for 100% of the equipment oil, the pit shall be connected to the holding tanks of sufficient capacity, through inter-connected pipes, where free flow of oil/water is possible from the pit to the holding tank.

ii. Holding Tanks

Water and oil may be piped from a pit to a holding tank or pond where the oil is separated naturally by gravitational force. The oil is then removed with a skimmer.

All materials used in the construction of pits, drainage pipes and holding tanks shall be non-combustible materials that are capable of withstanding the thermal shock of the oil fire followed by the cooling effect of the water. They shall have adequate mechanical strength to withstand any traffic or equipment handling loads to which they may be subjected. The retention pit/catch basin, holding tanks and piping shall be sized to handle the maximum amount of oil available and the quantity of water that may be used in fire suppression system.

b. Separation

The separation of oil-filled transformers from other equipment and buildings shall be in accordance with the following guidelines:

i. Separation of Large Transformers from Buildings

Transformers containing 1890 liters or more of insulating oil shall be at a minimum distance of 3m from any building, in addition to the fire rated coating given to the wall as per clause 3.3.1.

ii. Separation Between Large Transformers

Large oil-filled transformers of rating 5 MVA and above, shall be separated by a concrete fire wall, having a minimum 4hour fire resistance rating. The transformer shall be separated from the wall with at least 3m of clear space. In special cases where 3m clearance is not possible due to space constraint, the clearance shall be as defined in tender document.

iii. Firewalls

The height of a fire wall shall not be less than 0.3m above the height of the transformer tank and its oil conservator (if applicable), transformer bushings, pressure-relief vents.

The fire wall shall extend at least 0.6m horizontally beyond the line of sight between all points on adjacent transformers. The height of the fire wall shall not be less than that required to break the line-of-sight from any point on the top of the transformer tank and its oil conservator (if applicable) to any adjacent transformer bushing and surge arrester mounted on the transformer.

iv. Separation Between Station Transformers

For oil filled auxiliary/grounding transformers with less that 1890 liters (500 gallons) of insulating oil and where a firewall is not provided, should be separated by a minimum of 1.5m from each other and adjacent structure/building.

c. Mechanical Protection of Outdoor Transformers

The protection of outdoor transformers in substations when installed in exposed location shall comply with HCIS references listed in Appendix I for each Security Directives (SEC).

d. Transformer Fire Detection and Alarm Systems

The fire detection and alarm system for the outdoor power transformers shall consist of cross-zoned "rate anticipation heat detectors" and shall be rated as per NFPA requirements, listed for outdoor application.

The installation of all components of fire detection and alarm system shall be in accordance with the requirements of NFPA 72 and TES-B-106.01.

3.10.3 Oil-Filled Capacitors

Power capacitors containing an appreciable amount of flammable liquid shall be installed outdoors with fence enclosure or on elevated structure of at least 2.4m high above adjoining ground level for personnel safety to prevent accidental contact with exposed energized parts, terminals or buses associated with them. They shall be located minimum 3 meters away from any building, not of fire resistive construction. In special cases where 3m clearance is not possible due to space constraint, the clearance shall be as defined in tender document.

3.10.4 Oil-Filled Reactors

Reactors containing an appreciable amount of flammable liquid shall be located outdoors with fence enclosure to protect the personnel from accidental contact with energized parts. The installation of oil-filled reactors shall comply with Article 470 of NFPA 70.

3.11 Cables

3.11.1 General

The selection and installation procedures of power and control cables in substations shall comply with the guidelines outlined in TES-P-104, TCS-P-104, TES-P-119.20 and TES-P-119.29. The cables used in substations should not propagate a fire.

Cables which are protected with passive protection, mentioned in the preceding clauses, active fire protection such as water spray system is not required.

3.11.2 Cable Installation & Practices

a. Underground Conduit System is the preferred method of installation for critical power and control cables. Critical power and control cables are cables whose loss would render the emergency shutdown and the fire protection or alarm systems inoperative.

Cable installations subject to vehicular load, such as road crossings, shall be in concrete encased (either steel reinforced or any other suitable reinforcement method) ducts or conduits.

The pull boxes shall be located and constructed in a manner as to prevent the entry of flammable liquids that can possibly ignite the cables.

Consideration should be given to the installation of fire stops where the cables enter and leave pull boxes, and where the cables leave the conduit system.

b. Cable Trenches shall be constructed in a manner as to prevent the entry of any combustible liquid, such as insulating oil, that can possibly ignite the cables.Trench covers shall be made of fire retardant material as appropriate for location.

   Fire stops or fire breaks shall be installed when the trench enters into the building and/or from one room to the other room or from one floor to the other floor.

c. Cable Basement or Cable Entry Room and Tunnels Walkthrough are normally used where there is a large number of cables. If the cables used are flame retardant, the separation distances given in ANSI/IEEE 384 are acceptable. If the cables are not flame retardant, or the proper separation cannot be achieved, a fire resistive barrier or shield shall be used between the trays or a fire retardant coating paint, white color, rated for 3 hours or approved equivalent shall be applied to the cables.

   Note: The cable tunnel shall be provided with partition and door at every 60 meters interval.

d. Cable Trays and Cable Supports are constructed normally in cable tunnels, basement and trenches for proper cable laying arrangement.When a cable tray passes through a firewall or vertically through building floors, the cables shall be sealed with multiple cable transit (MCT) or approved equivalent. Circuits in the cable spreading area should be limited to control and instrumentation functions, and to those power supply circuits and facilities serving the control room and instrument systems. Power supply feeders to equipment and control room distribution panels shall be installed in conduit or enclosed metallic raceways or other suitable enclosure.

   Cable penetration fire stops shall be installed where sleeves or tray penetrations are used beneath control panels or other panels.

   Special attention shall be given to the installation of fire stops on vertical cable runs since they will propagate fire more readily than horizontal runs.

3.12 Miscellaneous Fire Hazards

3.12.1 Batteries

The storage batteries installation shall be in accordance with TES-P-103.04.

3.12.2 Combustible Materials

The use of combustible materials shall be avoided in the selection of desks, filing cabinets, storage boxes, display boards, building insulation, air conditioning duct insulation and mounting boards.

3.12.3 Heating Equipment

Heating equipment such as space heaters, open type room heaters, etc., shall not be used in the control buildings unless the design, installation and electrical supplies are deemed necessary. Clearances for heat producing appliances shall be adequate to minimize the risk of igniting adjacent combustible materials.

3.12.4 Fire Hazard of Cables

3.12.4.1 Low Pressure Oil Filled (LPOF) Cables

An LPOF cable fault resulting from insulation breakdown can rupture the cable sheath thereby releasing the pressurized oil. A high fault arcing that follows the insulation breakdown can ignite or burn the oil. Since the oil in the LPOF cable system is maintained under pressure, burning oil can spread over a large area creating intense fire and heavy damage to neighboring substation equipment or facilities.

To safeguard against fire hazard, LPOF cables used to feed indoor GIS in substations shall be installed and terminated outdoors including oil reservoirs and associated devices. The oil/SF6 cable termination shall be of leak-proof sealing end design.

3.12.4.2 PVC and XLPE Cables

When PVC or XLPE insulated cable is exposed to fire, the insulation may burn and can generate smoke and corrosive fumes. Smoke can obliterate escape routes and could impair the ability to fight the fire. When materials such as PVC burns, chlorine is driven off which when combines with water forms hydrochloric acid. This acid can attack the contact or relays, electronic circuits and wiring terminations, and can prolong the substation restoration time.

Where emission of corrosive fumes in any fire may damage electrical/electronic equipment, or present a health risk, XLPE insulated cable is preferable although it burns more readily than PVC. PVC is inherently flame retardant but emits acid fumes when it burns.

3.12.4.3 Oil/SF6 Cable Termination (Sealing End)

In 380kV GIS substation facilities/buildings the Oil/SF6 cable termination (sealing end) shall be located 3m away from any building, not of fire resistive construction. In special cases where 3m clearance is not possible due to space constraint, the clearance shall be as defined in tender documents.

3.12.5 Fire Hazards for Oil Filled Equipment

Due to the high temperatures, that can be produced by an electrical fault, or by an external fire that may engulf oil-filled equipment, the insulating oil may cause a fire hazard when provided in electrical equipment. Hence, every possible means shall be taken into account to install outdoor oil-filled equipment away from other equipment and substation building.

4.0 ADDITIONAL REQUIREMENTS

When specified in project SOW/TS, general hydrant system for Indoor Substation outside areas shall be provided as mentioned below:

The water supply for general hydrant system shall be from reliable fire protection supplies. Fire pumps system shall be used for 380kV S/S and below.

A reliable method of replenishing the supply shall be provided. An underground concrete tank suitable for the system is desired, in accordance with appropriate National Grid Saudi Arabia specifications and SAF-HCIS requirements.

The underground water tank capacity shall be sized for minimum one hour duration.

The bidders shall submit the suitable calculations for Fire Pump system and sizing of the underground water tank for the satisfactory fire protection of the areas under protection.

4.1 Piping System

Steel Pipes exposed above ground and in concrete trench shall be galvanized type, Schedule 40, ASTM A53/*A53M* or equivalent and exposed parts painted red.

4.2 Fire Pumps

Pumps shall be listed for fire protection use and chosen on the basis of water supply available. The electrically driven fire pumps shall be connected to a reliable power source.

4.2.1 Centrifugal fire pumps

Horizontal pump with positive suction (electric and diesel split case) shall be used to supply the pressure to drive the water from the water tank through a low level located specific gravity valve. Horizontal pump shall comply with standard 90-TMSS-02.
Each of the fire pumps and their respective drive shall be separate and independent independent units. Each pump shall be capable of delivering not less than 150% of its rated capacity at a discharge pressure of not less than 65% of the rated head at the rated capacity.

Each pump shall be capable of delivering the water demand on the largest single fire risk and an additional of not less than 25 Qgpm hose allowance. Pump capacity shall be determined through hydraulic calculation on this basis. The shut-off pressure of each pump shall not exceed 120% for split case pump.

Horizontal fire pumps shall be used when positive suction head is available and vertical turbine pumps shall be used when positive suction head is not possible due to space constraints. Negative suction for horizontal fire pumps is not allowable in any case. Fittings and accessories shall be complete as recommended by NFPA 20.

4.2.2 Jockey Pump

Pressure Maintenance Pump (PMP) shall be electric motor driven and with a rated capacity of 5% Fire Pump Capacity in GPM. The controller shall start the jockey pump when the fire water main pressure drops 10% and stop the pump when the pressure exceed the normal pressure by 10%.

4.3 Controllers and Drivers

All controllers and pump drivers shall be listed. Controllers shall be marked "Fire Pump Controller" and shall show plainly the name of the manufacturer, the identifying designation and the complete power ratings. All electrical connections shall be supervised.Y-Delta magnetic starter shall be used.

4.4 Fire Hydrants & Hose Cabinets

There shall be individually controlled hydrant valves (7 bars minimum pressure) fixed in suitable operational positions at ground level enabling all points of the substation to be covered. The minimum number of hydrant valves to be provided shall be four. The hydrants (150mm diameter) shall have 2 x 65mm.dia outlets and one 100mm diameter pumper connection, model Kennedy K-81 or equivalent. Further, this hydrant shall be fitted with cast iron end caps with blind ends, the caps being attached to the valves with an approved substantial chain. One female instantaneous coupling shall be provided.Spacing between 2 hydrants shall be not more than 70 m.

Near each hydrant, one Hydrant hose cabinet finish in signal red shall be provided with the following accessories (one each required).

a. Adjustable type fire hose nozzle, 65 mm diameter

b. Crow bar

c. Fire Axe

d. Adjustable Hydrant wrench

e. Hose 2 x 65mm dia., 15m length with instantaneous coupling

Note: Hose in abovementioned cabinet shall be of best quality light synthetic fiber mildew and bacterial attack, and shall offer protection against acids, alkalis & oils. Each hose shall be supplied with light alloy instantaneous couplings in accordance with the requirements of BS 336. Provide adequate adaptors to connect instantaneous couplings for connection with NH threaded hoses

The hose cabinet shall be painted with two coats of red paint and marked 'FIRE HOSE CABINET' in white block letters.

5.0 SYSTEM TESTS & ACCEPTANCE

Upon completion of work and prior to provisional and final acceptance of the installations of any system, the system shall be subjected to all tests required by the appropriate NFPA standards, in the presence of ISD Representative.

The actual heat test of at least one thermostat detector per protected area, aside from the recommended test procedure by the manufacturer for transformer fire protection system, shall be conducted.

6.0 TABLE 1- TYPES OF DETECTING AND ACTUATING MEANS AND TYPE OF PROTECTION SYSTEMS FOR ALL SUBSTATION VOLTAGE LEVELS, 380 kV AND BELOW AT VARIOUS AREAS

The type of fire detecting and actuating means for the various areas of the substations and the required type of protection shall be in accordance with the table below:

FIRE DETECTION AND PROTECTION SYSTEMS AND FIRE PREVENTION REQUIREMENTS FOR **NATIONAL GRID SAUDI ARABIA** SUBSTATIONS		
Area	Type of Detecting and Actuating Means	Type of Protection for Substation Facility Classification: "1, 2 and 3"
Offices	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	CO_2 Portable extinguishers
Kitchen or Tea room	Fire Detection and Alarm System. Fixed temperature rated 57 ºC.	CO_2 Portable extinguishers
Workshops	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	CO_2 Portable extinguishers
Store room	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	Dry chemical portable extinguishers
Hallway, Corridor, Lobby & Stair Cases	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	CO_2 Portable extinguishers
GIS Hall (if any)	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio. If height of the hall is above 9.0 meters, optical beam smoke detectors shall be provided.	CO_2 Portable extinguishers
Pump room (If any)	Fire Detection and Alarm System. Bulb type sprinkler rated 57 ºC.	Wet Sprinkler Systems and CO_2 Portable extinguishers

Table 1 (Continued)

FIRE DETECTION AND PROTECTION SYSTEMS AND FIRE PREVENTION REQUIREMENTS FOR NATIONAL GRID SAUDI ARABIA SUBSTATIONS		
Area	Type of Detecting and Actuating Means	Type of Protection for Substation Facility Classification: "1, 2 and 3"
Oil Filled Transformer Unit	Fire Detection and Alarm System. Actuation shall consist of cross-zoned rate of anticipation thermal heat detector weatherproof type.	Passive Fire protection systems (Adequate separation, fire rated walls, sump pit, fire stops & others). Wheeled & Portable type Fire Extinguishers, class ABC or Water Spray Fixed System if condition not met as per clause 3.8.6.5.
Oil Tank Room	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	CO_2 portable Fire Extinguishers
GIS room (69kV to 380kV)	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio. In 69kV Air Insulated Switchgear room, each cubicle, by one (1) multi-sensors (combined optical and heat detectors) smoke detector.	CO_2 portable Fire Extinguishers for the room /Hall. Protect cubicle panels by an approved Clean Agent Fire Extinguishing System (UL – Listed System)
MV Switchgear rooms (13.8kV to 34.5kV)	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	Total Flooding with an Approved Clean Agent system and CO_2 portable Fire Extinguishers
Communication room	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	Total Flooding with an Approved Clean Agent and CO_2 portable Fire Extinguishers
Control room	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	Total Flooding with an Approved Clean Agent and CO_2 portable Fire Extinguishers
Cable Entries & Basement /Tunnels (13.8kV, 33kV, 34.5kV, 110kV, 115kV, 132kV, 230kV, 380kV)	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	Dry chemical portable Fire Extinguishers and as per clause 3.11.

Table 1 (Continued)

FIRE DETECTION AND PROTECTION SYSTEMS AND FIRE PREVENTION REQUIREMENTS FOR NATIONAL GRID SAUDI ARABIA SUBSTATIONS		
Area	Type of Detecting and Actuating Means	Type of Protection for Substation Facility Classification: "1, 2 and 3"
Battery Rooms	Combination of intrinsically safe and explosion proof flame and heat detectors plus hydrogen gas detectors.	Total Flooding with an approved clean agent and CO_2 portable Fire Extinguishers Dry chemical Fire Extinguishers installed outside near the door, See clause 3.8.6.8.
HVAC Equipment (Mechanical room)	A combination of smoke and heat detectors in 1:1 Ratio.	Dry chemical portable Fire Extinguishers
HVAC Main Supply and Return Ducts	Approved photo-optical type duct smoke detector.	
AC/DC Distribution Room	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	Total Flooding with an Approved Clean Agent and CO_2 portable Fire Extinguishers
SCADA and Relay Rooms	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	Total Flooding with an Approved Clean Agent and CO_2 portable Fire Extinguishers.
Toilets	Rate of rise & fixed temperature heat detector.	Dry chemical portable Fire Extinguishers
Security Gate Houses	Multi-sensors (combined optical and heat detectors) or optical type of smoke detector.	Dry chemical class ABC Fire Extinguishers
Clean Agent cylinder room	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	CO_2 portable Fire Extinguishers

Table 1 (Continued)

FIRE DETECTION AND PROTECTION SYSTEMS AND FIRE PREVENTION REQUIREMENTS FOR NATIONAL GRID SAUDI ARABIA SUBSTATIONS		
Area	Type of Detecting and Actuating Means	Type of Protection for Substation Facility Classification: "1, 2 and 3"
Station Transformer	Fire Detection and Alarm System. Adequate separation, fire rated walls, sump pit, fire stops.Weatherproof rate anticipation heat detector.	Dry chemical portable Fire Extinguishers and as per clause 3.10.2.b.
Shunt Capacitor	Fire Detection and Alarm System. Adequate separation, fire rated walls, sump pit, fire stops.Weatherproof rate anticipation heat detector.	Dry chemical portable Fire Extinguishers
Shunt Reactor	Fire Detection and Alarm System. Adequate separation, fire rated walls, sump pit, fire stops.Weatherproof rate anticipation heat detector.	Dry chemical portable Fire Extinguishers
Record Room	Fire Detection and Alarm System. A combination of optical and multi-sensors (combined optical and heat detectors) smoke detectors in 1:1 ratio.	Dry chemical portable Fire Extinguishers

Note: Both **Novec 1230** & **FM200** are approved clean agents as per SAF -04 of HCIS.

7.0 APPENDIX I

HIGH COMMISSION FOR INDUSTRIAL SECURITY (HCIS),
MINISTRY OF INTERIOR

LIST OF SECURITY DIRECTIVES FOR INDUSTRIAL FACILITIES
(Issue Date: 26/05/2010G)

HCIS SEC REFERENCE NO.	DESCRIPTION
SEC-01	Application of Security Directives
SEC-02	Security Fencing
SEC-03	Security Gate
SEC-04	Security Lighting
SEC-05	Integrated Security System
SEC-06	Security Devices
SEC-07	Power Supply
SEC-08	Communications
SEC-09	Security Doors
SEC-10	Security Locks
SEC-11	Identification Cards
SEC-12	Information Protection

Note:

PDF copies of aboveSEC directives are not available on the webpage of National Grid Saudi Arabia Standards and Specifications Department.

The verbatim copies of above Security Directives (SEC) of HCIS are availabe and procurable from Industrial Security Performance Department / Security Development Division.
E-mail Address: IPDSDDV@se.com.sa

8.0 APPENDIX II

HIGH COMMISSION FOR INDUSTRIAL SECURITY (HCIS) ,
MINISTRY OF INTERIOR

LIST OF SAFETY AND FIRE PROTECTION DIRECTIVES FOR INDUSTRIAL FACILITIES
(Issue Date: 26/05/2010G)

HCIS SAF REFERENCE NO.	DESCRIPTION
SAF-01	Application of Safety and Fire Protecion Directives
SAF-02	Enviromental, Health and Safety (EHIS) Management
SAF-03	Plant Buildings
SAF-04	Fire Protection Systems
SAF-05	Industrial Drainage
SAF-06	Plant Layout, Spacing and Access
SAF-07	Wellsite Safety
SAF-08	Storage Tanks and Storage Vessels
SAF-09	Fireproofing in Plants
SAF-10	Pressure Piping, Pressure Vessels and Transportation Pipelines
SAF-11	Emergency Shutdown, Isolation and Depressuring
SAF-12	Electrical Safety
SAF-13	Work Permits
SAF-14	Safe Manufacture, Transportation, Storage and Use of Explosive Materials and Pyrotechnic Articles
SAF-15	Private Industrial Fleet Vehicle Filling Stations
SAF-16	Bulk Plants and Air Fueling Support Facilities
SAF-17	Offshore Production Facilities
SAF-18	Mines and Mineral Processing Plants
SAF-19	Electric Power Generating Plants and Associated Facilities
SAF-20	Pre-Incident Planning and Management of Emergencies
SAF-21	Personal Protective Equipment (PPE)
SAF-22	Portable Electric and Electronic Devices

Note:

PDF copies of aboveSAF directives are not available on the webpage of National Grid Saudi Arabia Standards and Specifications Department.

The verbatim copies of above Safety and Fire Protection Directives (SAF) of HCIS are availabe and procurable from Industrial Security Performance Department / Security Development Division.
E-mail Address: IPDSDDV@se.com.sa

9.0 APPENDIX III: SUBSTATION FACILITY CLASSIFICATION

The classification criteria of NATIONAL GRID SAUDI ARABIA facilities under the supervision of the High Commission for Industrial Security (HCIS) are as follows:

First category: "Class 1 facility"

No.	Criteria
1	Direct impact on oil production and oil exportation in case of loss
2	It affects the stability of national security (That feeds government and essential military facilities)
3	Very important for the electrical grid in case of loss. (can't be compensated from another substation)

Second category: "Class 2 facility"

No.	Criteria
1	Direct impact on oil production with no effect on oil exportation in case of loss
2	Moderately important for the electrical grid in case of loss. (can be compensated partially)

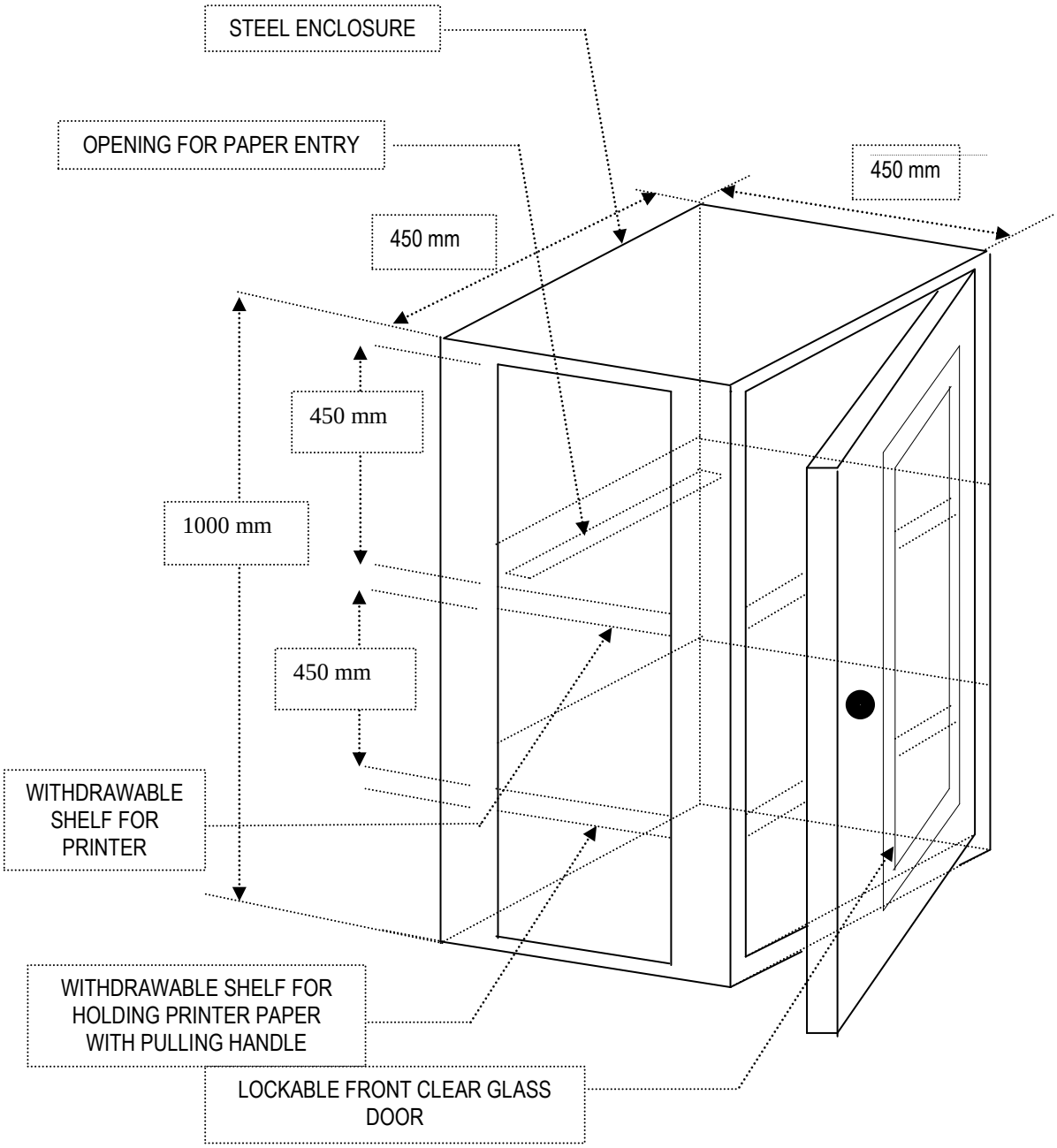
Third category: "Class 3 facility"

No.	Criteria
1	No impact on oil production and oil exportation in case of loss
2	No impact on the stability of national security in case of loss
3	Of little importance for the electrical grid in case of loss. (can be compensated from another substation)

Fourth category: "Class 4 facility"

Note 1:	This facility is considered as a remote support facility to Substation with Category "1, 2 & 3" only (i.e remote warehouse & office support facility)
Note 2:	Transmission Substations shall not be classified under this category as this is not applicable.

10.0 DWG # TE-1921-0100-00



NOTES

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. PRINTER POWER SUPPLY AND DATA CABLE SHALL ENTER FROM THE BACK SIDE OPENING. THERE SHALL BE OPENING IN THE TOP SHELF FOR PAPER ENTRY.

Figure 21-1 : Steel Enclosure with Withdrawable Shelf for Fire Alarm Printer and Paper

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11.0 BIBLIOGRAPHY

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2. 71-TMSS-03, Normal Weight Ready-Mixed Portland Cement Concrete
3. 90-TMSS-01, Fire Protective signaling Panel
4. 90-TMSS-02, Fire Water Pumps
5. TCS-P-104, Underground Cable Construction Standards
6. TES-B-106.01, Fire Alarm Systems
7. TES-B-106.03, CO_2 Total Flooding systems
8. TES-B-106.05, HFC-227ea Clean Agent Extinguishing System
9. TES-B-106.06, FK 5-1-12, Fixed Installation Total Flooding Clean Agent Fire Extinguishing System
10. TES-H-107.02, Paint Color Codes and Standards
11. TES-K-100.01, Heating, Ventilating and Air Conditioning System
12. TES-P-119, Substation Design Standards
13. TES-P-104, Underground Cable Engineering Standards
14. TES-P-103.04, Storage Battery Installation
15. ANSI/IEEE C2, National Electrical Safety Code
16. ANSI/IEEE 383, IEEE Standard for Qualifying Class 1E Electric Cables and Field Splices for Nuclear Power Generating Stations
17. ANSI/IEEE 384, IEEE Standard Criteria for Independence of Class 1E Equipment and Circuits
18. ANSI/IEEE 980, IEEE Guide for Containment and Control of Oil Spills in Substations
19. ANSI/IEEE 979, Guide for Substation Fire Protection
20. ASTM E814, Standard Test Method for Fire Tests of Through Penetration Fire Stop
21. ASTM A53/A53M, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
22. NFPA 10, Standard for Portable Fire Extinguishers

23. NFPA 12, Standard or Carbon Dioxide Extinguishing Systems

24. NFPA 13, Standard for the installation of Sprinkler Systems

25. NFPA 15, Standard for Water Spray Fixed Systems for Fire Protection

26. NFPA 20, Standard for the installation of Stationary Pumps for Fire Protection

27. NFPA 70, National Electrical Code

28. NFPA 72, National Fire Alarm Code and Signaling Code

29. NFPA 80, Standard for Fire Doors and-Other Opening Protectives

30. NFPA 90A, Standard for the Installation of Air Conditioning and Ventilating Systems

31. NFPA 101, Life Safety Code

32. NFPA 220, Standard on Types of Building Construction

33. NFPA 221, Standard for High Challenge Fire Walls, Fire Walls and Fire Barrier Walls

34. NFPA 251, Standard Methods of Tests of Fire Resistance of Building Construction and Materials

35. NFPA 255, Standard Method of Test of Surface Burning Characteristics of Building Materials

36. NFPA 850, Recommended Practice for Fire Protection for Electric Generating Plants and High Voltage Direct Current Converter Stations

37. NFPA 2001, Standard on Clean Agent Fire Extinguishing Systems

38. BS 336, Specification for fire hose couplings and ancillary equipment

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42. UL 555S, Standard for Safety Smoke Damper

43. UL 1479, UL Standard for Safety Fire Tests of Through Penetration Fire Stops